Jinxi (Jace) Yu

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RESEARCH INTERESTS

My research focuses on **LLM reasoning, multi-agent systems, and efficient inference**. I study structured memory, planning, verification, and adaptive collaboration for reliable long-horizon reasoning. I am interested in how agent architectures, communication topologies, and inference-time compute allocation improve reasoning quality and efficiency, enabling scalable, trustworthy systems for complex tasks.

EDUCATION

University of California, Los Angeles (UCLA) Los Angeles, CA
B.S. in Computer Science *Expected Jun 2027*
GPA: 3.97/4.00 | **Honors:** Dean's Honors List
Relevant Coursework: ML, NLP, DL, Object-Oriented Programming (C++), Algorithms, Operating Systems, Data Mining.

PUBLICATIONS & PREPRINTS

[1] **Jinxi Yu**, Eric Hanchen Jiang, Junyi Zhang, et al. **Continual Graph Memory and Community Infrastructure for AI-Assisted Mathematical Discovery.** Under review at **Nature Machine Intelligence**. arXiv:2610.02945

[2] **Jinxi Yu**, Eric Hanchen Jiang, Levina Li, Dong Liu, Zhi Zhang, Wenxiao Zhao, Yanxuan Yu, Kai-Wei Chang, and Ying Nian Wu. **Privacy-Preserving Topology-Guided Safety for LLM-Based Multi-Agent Systems via Federated Graph Learning.** Under review at **AAAI 2027**. arXiv:2609.02967

[3] **Jinxi Yu**, Yubei Li, Eric Hanchen Jiang, Zhi Zhang, Dong Liu, Wenxiao Zhao, Levina Li, Kai-Wei Chang, and Ying Nian Wu. **Codebook Agent: Amortized Topology Design for LLM Multi-Agent Systems.** Under review at **AAAI 2027**. arXiv:2609.02264

[4] **Jinxi Yu**, Pranav Subbaraman, Fang Sun, Yue Yao, Huacong Tang, Xiao Luo, and Yizhou Sun. **Accelerating Time Series Foundation Models with Speculative Decoding.** Under review at **AAAI 2027**. arXiv:2511.18191

[5] Eric Jiang, Xiao Liang, Junyi Zhang, Justin Baker, Zhi Zhang, Yingjia Wan, Yubei Li, **Jinxi Yu**, et al. **From Solvers to Research: Large Language Model-Driven Formal Mathematics at the Research Frontier.** Under review at **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**. arXiv:2607.07779

RESEARCH & WORK EXPERIENCE

UCLA Moonshot Math Harness Jul 2026 – Present
Research Assistant | Advisors: Prof. Kai-Wei Chang, Prof. Nanyun Peng, Prof. Terence Tao

- Built a **long-horizon multi-agent reasoning harness** for autonomous mathematical research, orchestrating explorer, verifier, and consultant agents through a library of skills and Model Context Protocol (MCP) tools. Enforced role-specific capabilities in the harness.
- Designed a **persistent graph-based memory system** to preserve intermediate results, dependencies, and competing proof directions across long-running investigations. Developed **self-evolving memory** curation and iterative planning to prioritize promising branches and adapt problem-solving strategies for open mathematical problems [1].

UCLA Natural Language Processing Lab Mar 2026 – Present
Research Assistant | Advisors: Prof. Kai-Wei Chang, Prof. Ying Nian Wu

- **Privacy-preserving agent safety [2].** Trained an edge-featured graph attention network over inter-agent communication graphs through federated learning without sharing raw traces. Outperformed centralized training on **three benchmarks** and cut the live attack-success rate by **43%** with near-unguarded utility, using a 0.5M-parameter guard with approximately 50 ms CPU latency.
- **Efficient multi-agent topology design [3].** Developed a **16-entry VQ-VAE codebook**, reward-weighted predictor, and execution-grounded MLP reranker. Achieved the best accuracy on all **six evaluated benchmarks** (84.6 average vs. 83.0 for the strongest prior method), **2.4 ms** topology generation (over 120× faster than graph-diffusion designers), and **21.9–33.2% fewer LLM tokens**.

UCLA SCAI Lab (AI and Data Mining) Feb 2026 – Present
Research Assistant | Advisor: Prof. Yizhou Sun

- Developed **Gaussian speculative decoding** in PyTorch with lightweight drafts and single-pass parallel verification, achieving up to **3.0**× speedup across **five time series foundation model families** without retraining. Proved a **capped-geometric throughput law** and a **bounded-deviation guarantee** on forecast fidelity [4].

Huatai Securities Co., Ltd. (Wealth Management Dept.) Nanjing, China
Data Analyst Intern Jun 2025 – Jul 2025

- Conducted quantitative analysis of **20+ institutional investment portfolios** across **5 years of historical data**, evaluating return distributions, volatility, maximum drawdowns, and risk-adjusted performance to identify investment trends.
- Developed **Python-based data analysis pipelines** to clean and process financial datasets, calculate portfolio performance metrics, and visualize asset allocation and historical returns to support investment research and decision-making.

TECHNICAL SKILLS

Programming: Python, C++, Java, Go, SQL, JavaScript, TypeScript, HTML/CSS.
ML / Research Tools: PyTorch, CUDA profiling, pandas.
Systems / Development: Git, Bash, Docker, AWS, Linux, MySQL, PostgreSQL, Redis, Node.js.